Sparse Recovery and Compressed Sensing

A Model-Based Deep Learning Approach

Core question: How far can model-based structure take us — and when do we need data-driven flexibility?

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Based on: Chen, Liu, Wang, Yin — "Hyperparameter Tuning is All You Need for LISTA" · NeurIPS 2021

Seminar Roadmap

1	2	3	4	5
Background	**Unrolling**	**Our Study**	**Results**	**Images**
Motivation, LASSO, ISTA / FISTA	LISTA, ALISTA, HyperLISTA	Scalar variants + training methods	Sparse recovery & design space	MNIST & FashionMNIST CS stress tests

Goal: systematically compare how much learning is needed when moving from ideal sparse signals to real compressed-sensing images.

Synthetic signals
m=250, n=500, s=50
Gaussian A, noiseless

Trained models
LISTA, ALISTA, HyperLISTA
+ 3 scalar designs (ours)

Image CS
MNIST & FashionMNIST pixel domain
ratios $\in \{0.25, 0.5\}$

Motivation & Problem Formulation

Sparse recovery appears everywhere:

Application	Measurements	Goal
MRI acceleration	k-space (m ≪ n)	Image pixels
Radar	Compressed snapshots	Reflectivity
Hyperspectral	Spectral bands	Spatial cube
Compressed sensing	Random projections	Original signal

Why tractable? Natural signals have only s ≪ n non-zero components.

LASSO — the optimization backbone:

$$\hat{x} = \arg\min_x \tfrac{1}{2} \| b - A x \|_2^2 + \lambda \| x \|_1$$

LASSO = Least Absolute Shrinkage and Selection Operator (Tibshirani, 1996)

Measurement model

$$b = A x^* + \varepsilon$$

A ∈ ℝ^{m×n}, m ≪ n

x^*: s-sparse signal

ε: additive Gaussian noise

Metrics: NMSE (dB) for synthetic signals;
NMSE + PSNR + SSIM for image CS.

Classical Baselines: ISTA and FISTA

ISTA — proximal gradient descent on the LASSO objective:

$$x^{(k+1)} = S_\theta\left(x^{(k)} + (1/L)\, A^T\left(b - A\, x^{(k)} \right) \right), \qquad L = \|A\|_2^2, \qquad \theta = \lambda / L$$

FISTA — ISTA with Nesterov momentum, $O(1/k^2)$ convergence vs $O(1/k)$:

$$y^{(k)} = x^{(k)} + \left((t_k - 1) / t_{k+1} \right) \left(x^{(k)} - x^{(k-1)} \right), \qquad x^{(k+1)} = S_\theta\left(y^{(k)} + (1/L)\, A^T\left(b - A\, y^{(k)} \right) \right)$$

Method	NMSE @ K=16	Error energy [%]	Params
ISTA	−5.3 dB	29.5%	0
FISTA	−11.0 dB	7.9%	0

At K=16 iterations both methods are far from converged.
ISTA needs 200+ iterations for −20 dB.

Can we make 16 steps count?

ISTA = Iterative Shrinkage-Thresholding Algorithm;   FISTA = Fast ISTA (Beck & Teboulle, 2009)

Deep Unrolling: From Iterations to Layers

Key idea (Gregor & LeCun, 2010): unfold K ISTA iterations into a K-layer network and learn the update parameters from data.

Classical ISTA — parameters fixed

$$z^{(k)} = x^{(k)} + (1/L)\, A^T(b - A\, x^{(k)})$$

$$x^{(k+1)} = S_\theta(z^{(k)})$$

Parameters 1/L and θ = λ/L are fixed constants.

Fixed — no learning

Unrolled LISTA — parameters learned

$$z^{(k)} = W_x^{(k)}\, x^{(k)} + W_y^{(k)}\, b$$

$$x^{(k+1)} = S_{\theta_k}(z^{(k)})$$

W_x(k), W_y(k), θ_k are learned per layer via BPTT.

~6 M parameters — learned end-to-end

MBDL principle: preserve the known algorithmic structure (gradient step + soft-threshold);
learn only what improves finite-depth reconstruction. Training minimizes $\|x(K) - x^*\|^2$ via BPTT.

BPTT = Backpropagation Through Time — end-to-end backpropagation through all K unrolled layers, as when training a recurrent network

The MBDL Ladder: LISTA → ALISTA → HyperLISTA

LISTA — learn all matrices per layer, ~6 M params:

$$x^{(k+1)} = S_{\theta_k}\left(W_x^{(k)}\, x^{(k)} + W_y^{(k)}\, b \right)$$

ALISTA — W analytic, learn step sizes and thresholds, 32 params:

$$W^* = \left(A\, A^T \right)^{-1} A \qquad \textit{(column-normalized)}$$

$$x^{(k+1)} = S_{\theta_k}\left(x^{(k)} + \gamma_k\, W^{*T}\left(b - A\, x^{(k)} \right) \right)$$

HyperLISTA — adaptive θ/β/p from residuals, 3 params (grid search, zero backprop):

$$\theta^{(k)} = c_1\, \mu\, \left\| A^+\left(A\, x^{(k)} - b \right) \right\|_1, \qquad \beta^{(k)} = c_2\, \mu\, \left\| x^{(k)} \right\|_0$$

$$p^{(k)} = c_3 \log\left(\left\| A^+ b \right\|_1 \Big/ \left\| A^+\left(A\, x^{(k)} - b \right) \right\|_1 \right)$$

Model	Params	NMSE	Error [%]
ISTA	0	−5.3 dB	29.5%
FISTA	0	−11.0 dB	7.9%
LISTA	6 M	−20.3 dB	0.93%
ALISTA	32	−29.8 dB	0.10%
HyperLISTA	**3**	**−61.4 dB**	**0.00007%**

NMSE on synthetic sparse recovery (m=250, n=500, s=50, noiseless, K=16 layers)

More structure → fewer params → better performance

This is the MBDL principle in action.

Our Design Space Study: What Should We Learn?

Research question: LISTA learns 6 M parameters; HyperLISTA uses 3 scalars. What is the minimal useful set of learnable components?
The ISTA gradient-correction update has two free scalar sequences:

$$x^{(k+1)} = S_{\theta_k}\left(x^{(k)} + \gamma_k A^T\left(b - A\, x^{(k)} \right) \right)$$

We designed three models, each learning a different subset:

ThresholdISTA

16 params

Learns $\theta_1, \ldots, \theta_K$ only

Step fixed at $\gamma = 1/L$

Init: $\theta_0 = \lambda/L$

StepISTA

16 params

Learns $\gamma_1, \ldots, \gamma_K$ only

Threshold coupled: $\theta_k = \lambda\gamma_k$

Init: $\gamma_0 = 1/L$

StepThresholdISTA

32 params

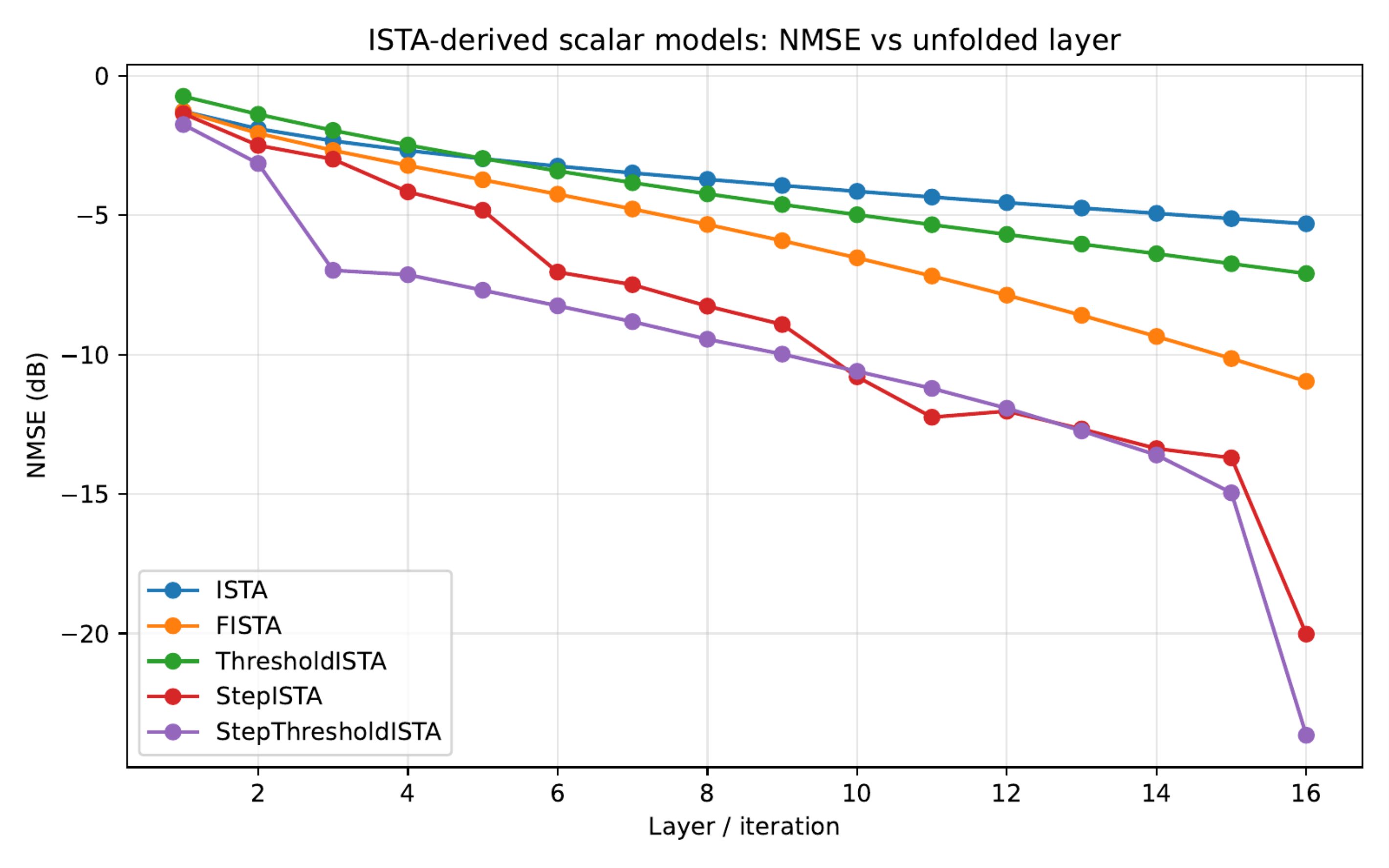
Learns both γ_k and θ_k

independently per layer

Init: ISTA values

All parameters softplus-reparameterized (positivity). All initialized to reproduce ISTA exactly before training.

Our Results: Scalar Models



Model	NMSE	Params
ISTA	−5.3 dB	0
FISTA	−11.0 dB	0
ThresholdISTA	−7.1 dB	16
StepISTA	**−20.0 dB**	**16**
StepThresholdISTA	**−23.6 dB**	**32**
LISTA-L1 (ref.)	−20.3 dB	6 M

Key finding

Learning γ_k alone (16 scalars) already matches full LISTA (6 M parameters).

Decoupling θ_k adds 3 dB at zero extra cost.

StepThresholdISTA beats LISTA-L1 with 187,000× fewer parameters.

Training Methods for Unfolded Optimizers

Architecture fixed (LISTA). Only the training strategy changes.

L1 — End-to-end

$$L = \| x^{(K)} - x^* \|^2$$

Supervise final output only.
Standard BPTT.

L2 — Deep Supervision

$$L = \|x^{(K)} - x^*\|^2 + \lambda \Sigma_k w_k \|x^{(k)} - x^*\|^2$$

Supervise every
intermediate layer.

L3 — Greedy Layerwise

$$\min_{\theta_k} \| x^{(k)} - x^* \|^2$$

Train layer k while
layers 1…k−1 frozen.

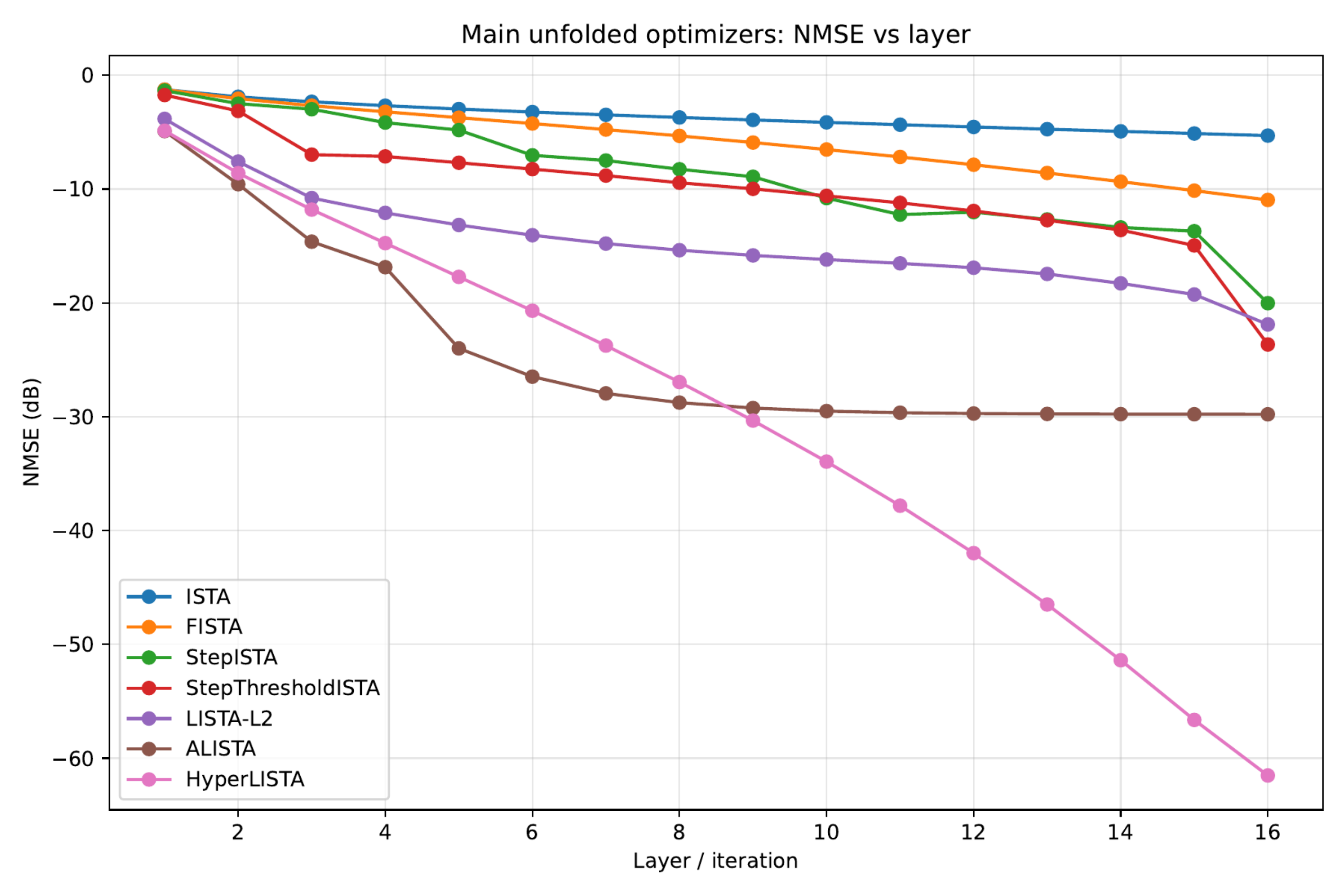
Tied vs. Untied weights:

	Untied LISTA	Tied LISTA
Weights	Independent per layer	Single shared (Wx, Wy, θ)
Params	~6 M	375 K
Style	Fully flexible	RNN-like
Effect	Hard to optimize	Smoother loss landscape

Weight tying = implicit regularization.

LISTA-Tied-L1: −19.1 dB @ 375 K
LISTA-L1: −20.3 dB @ 6 M

Synthetic Sparse Recovery Results

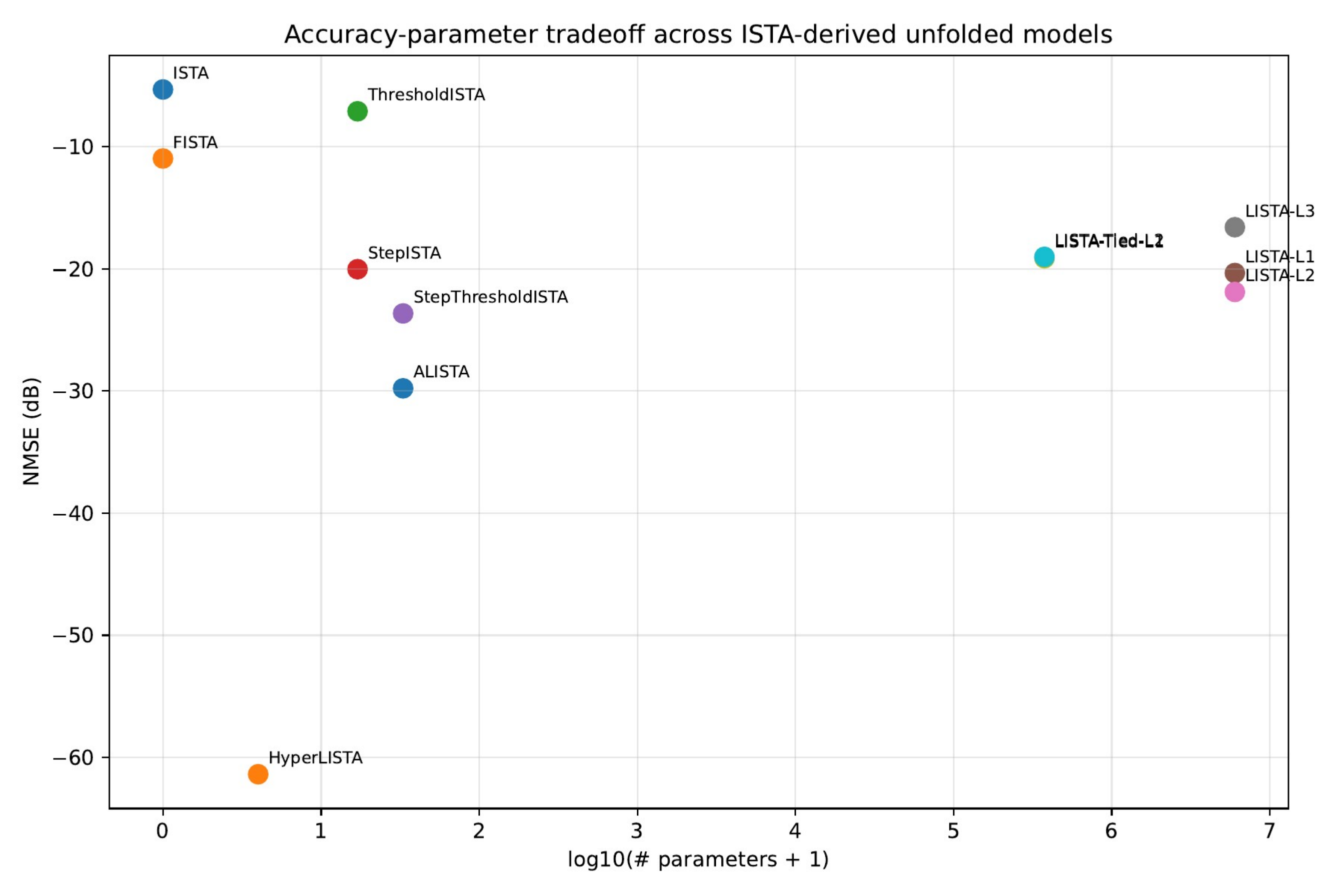


Method	NMSE	Params
ISTA	−5.3 dB	0
FISTA	−11.0 dB	0
LISTA-L1	−20.3 dB	6 M
LISTA-L2	−21.9 dB	6 M
LISTA-L3	−16.6 dB	6 M
LISTA-Tied-L1	−19.1 dB	375 K
ALISTA	**−29.8 dB**	**32**
HyperLISTA	**−61.4 dB**	**3**

L3 plateau: NMSE flattens ~layer 8.
Greedy training gives no incentive to improve beyond what prior layers solved.

LISTA-Tied > LISTA-Untied despite 16× fewer params.

Design-Space Map



Model	Params	NMSE	What drives it
ThresholdISTA	16	−7.1 dB	learned θ_k
StepISTA	16	−20.0 dB	learned γ_k
StepThresholdISTA	**32**	**−23.6 dB**	**γ_k + θ_k**
LISTA-L1	6 M	−20.3 dB	full matrices
ALISTA	32	−29.8 dB	analytic W
HyperLISTA	**3**	**−61.4 dB**	**adaptive residuals**

StepThresholdISTA (32 scalars)
beats LISTA-L1 (6 million matrices).
The step-size schedule γ_k is the bottleneck.

Real Image CS: FashionMNIST Pixel Domain

Setup: flatten 28×28 image to $x \in [0,1]$^{784}, measure $b = Ax$, ratio $m/784 \in \{0.25, 0.5\}$. FashionMNIST: ~51.5% of pixels exactly zero.

Method	NMSE	PSNR	SSIM	Params
ISTA	−1.2 dB	8.9	0.13	0
FISTA	−1.2 dB	8.9	0.14	0
StepThresholdISTA	−1.2 dB	8.9	0.13	32
ALISTA	−0.9 dB	8.4	0.12	32
HyperLISTA	−0.3 dB	7.7	0.09	3
LISTA-L2	**−13.7 dB**	**22.3**	**0.82**	**12 M**

ratio = 0.25

Signal model mismatch

All structured methods assume i.i.d.
Gaussian sparse signals.
FashionMNIST pixels are bounded [0,1],
spatially correlated, and only soft-sparse.

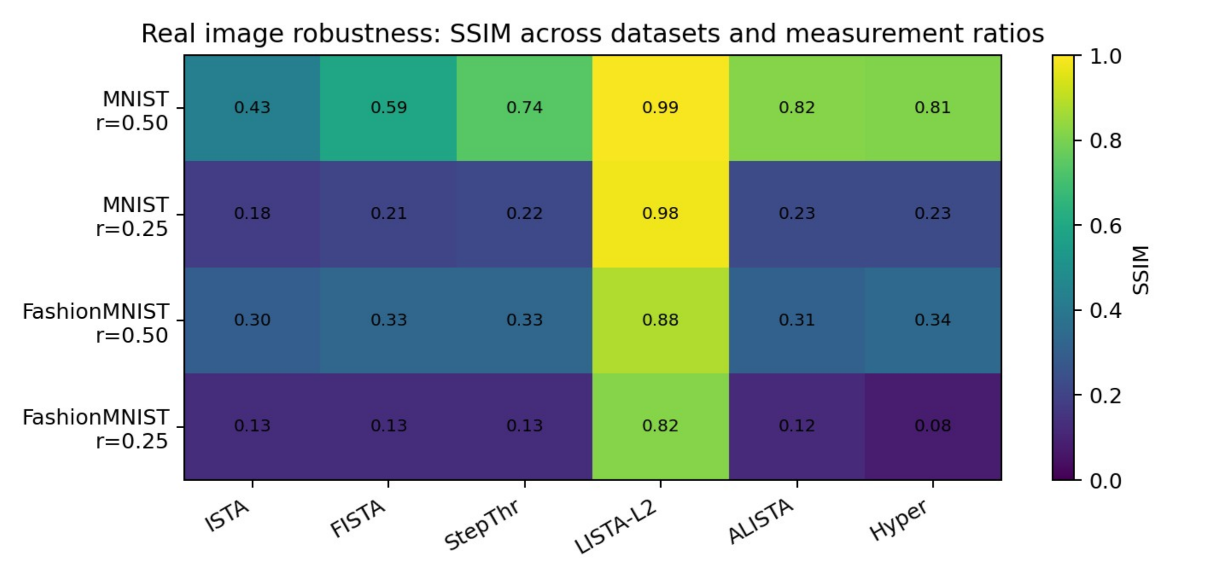
Why LISTA wins

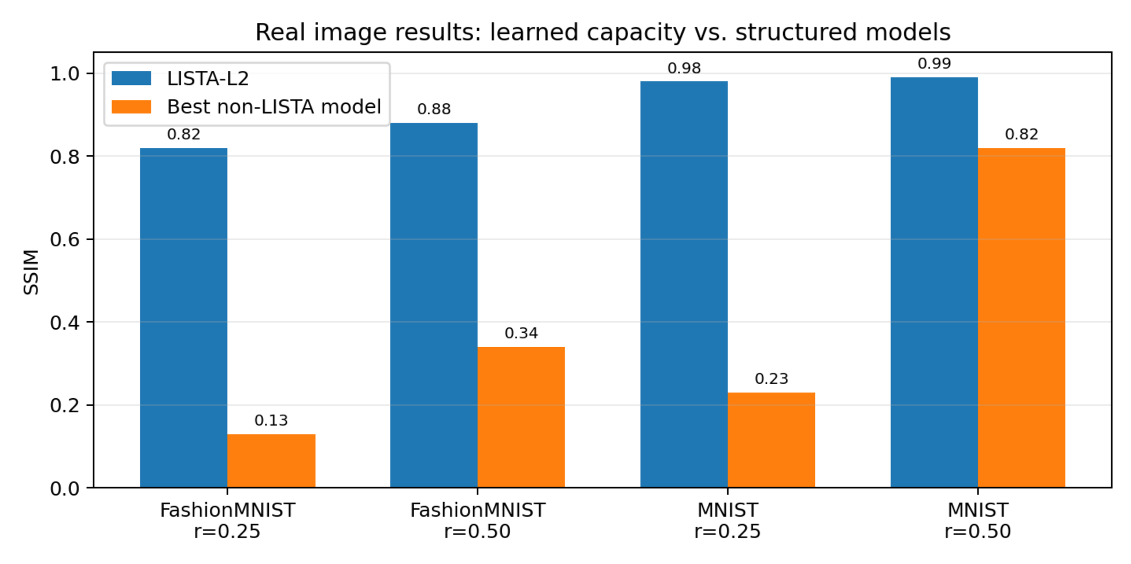
LISTA-L2 is assumption-free —
it learns the actual image distribution
from 51 K training images.

Wrong prior is worse than no prior.

Robustness Across Datasets and Measurement Ratios

SSIM across {MNIST, FashionMNIST} × ratio {0.25, 0.5}: harder data and fewer measurements widen the gap.





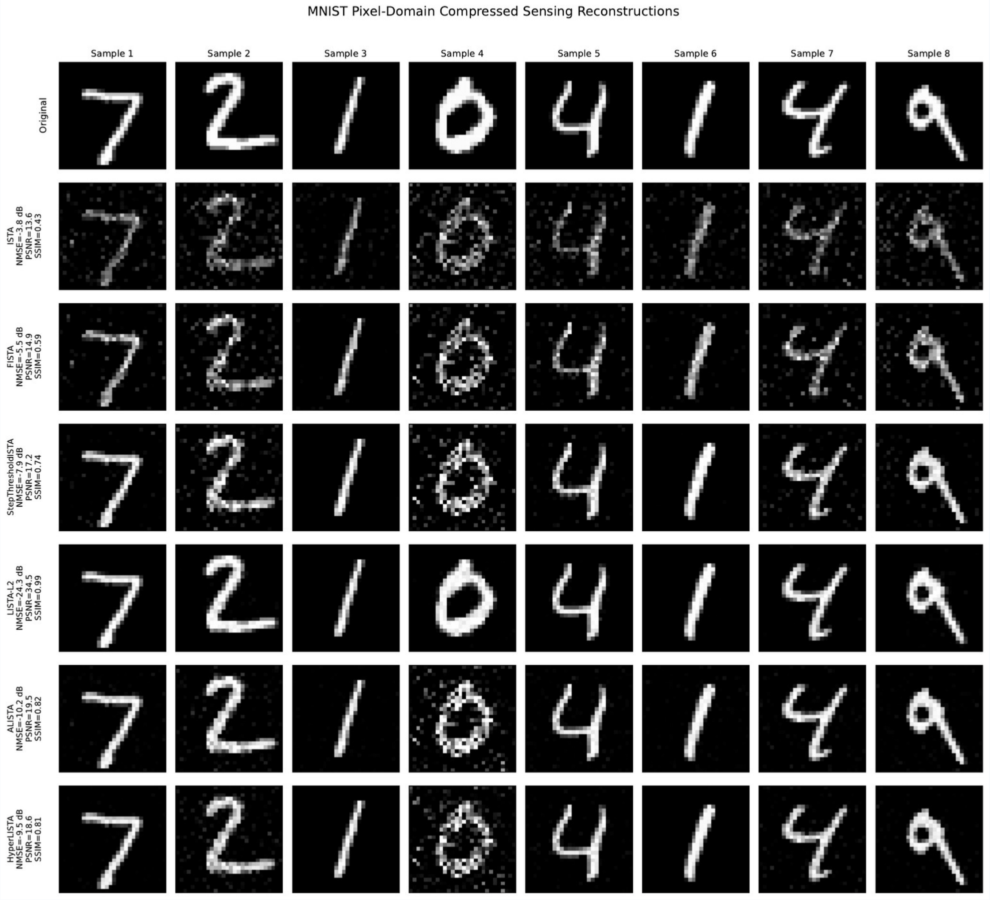
Trend

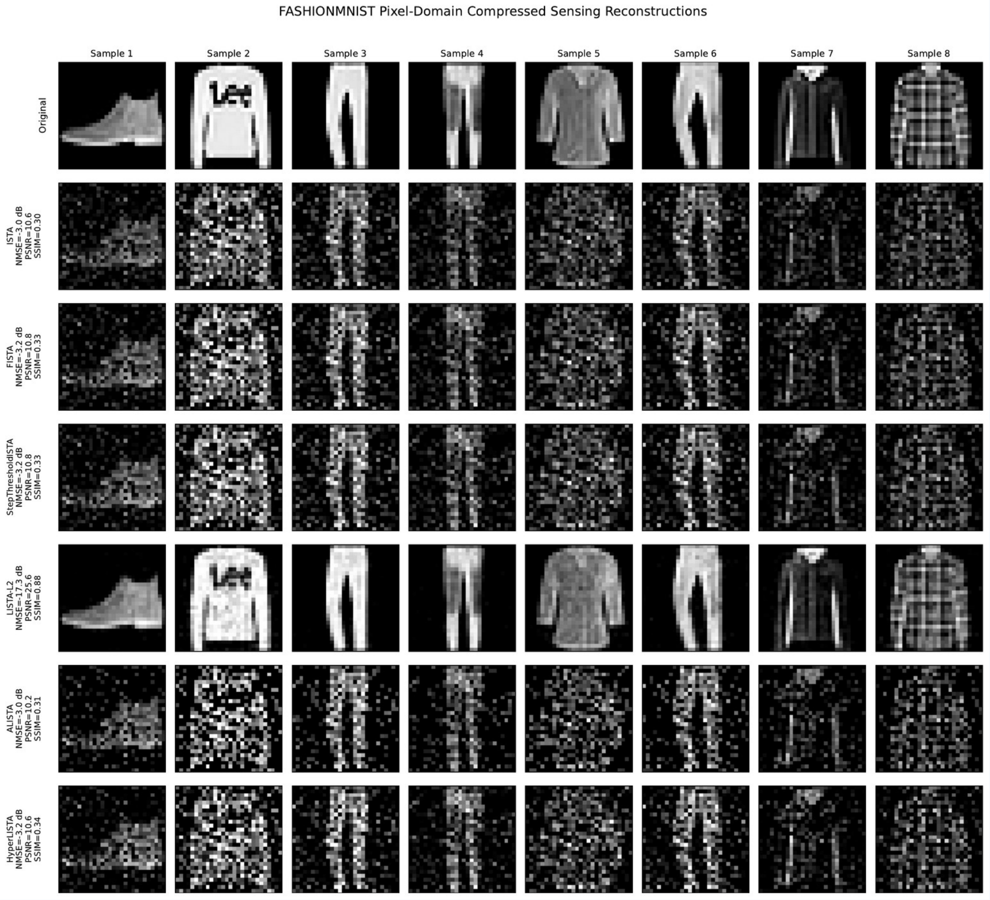
MNIST @ 0.5 is easy for everyone; FashionMNIST @ 0.25 breaks every structured model. LISTA-L2 stays ≥ 0.82 SSIM everywhere.

Interpretation

The sparse-Gaussian prior degrades gracefully on MNIST (~78% zeros) but fails on FashionMNIST (~51% zeros, textured).

Visual Stress Tests (ratio = 0.25)





LISTA-L2 recovers clean structure on both datasets. ALISTA / HyperLISTA zero out valid signal on FashionMNIST — the Gaussian-sparse threshold is miscalibrated for textured images. Wrong prior is worse than no prior.

Conclusions

1 Structure wins when assumptions match.

HyperLISTA's 3 scalars beat 6 M learned parameters by 41 dB on synthetic sparse data.

2 Step sizes matter most.

StepISTA (16 scalars) matches LISTA. Decoupling θ_k adds 3 dB. Matrix learning is largely redundant.

3 Weight tying is implicit regularization.

LISTA-Tied matches LISTA-Untied (−19.1 vs −20.3 dB) with 16× fewer parameters.

4 Wrong prior actively hurts.

ALISTA / HyperLISTA score below ISTA on real images. LISTA wins by making no structural assumptions.

MBDL hierarchy on synthetic data:
ISTA → LISTA → ALISTA → HyperLISTA
0 → 6M → 32 → 3 params
−5 → −20 → −30 → −61 dB

Future work

Apply models in DCT / wavelet domain — images are sparser there, closer to the Gaussian-sparse model.

Design image-aware thresholds for bounded, non-negative pixel structure.

Code & References

Code — public GitHub repository (notebooks 01–04 + src modules):

github.com/ThEpiCake/hyperlista-mbdl

References:

[1] A. Beck and M. Teboulle, "A Fast Iterative Shrinkage-Thresholding Algorithm for Linear Inverse Problems," SIAM J. Imaging Sciences, 2009.

[2] K. Gregor and Y. LeCun, "Learning Fast Approximations of Sparse Coding," ICML 2010.

[3] X. Chen, J. Liu, Z. Wang, W. Yin, "Theoretical Linear Convergence of Unfolded ISTA and Its Practical Weights and Thresholds," NeurIPS 2018.

[4] J. Liu, X. Chen, Z. Wang, W. Yin, "ALISTA: Analytic Weights Are As Good As Learned Weights in LISTA," ICLR 2019.

[5] X. Chen, J. Liu, Z. Wang, W. Yin, "Hyperparameter Tuning is All You Need for LISTA," NeurIPS 2021.

[6] N. Shlezinger and Y. C. Eldar, "Model-Based Deep Learning," Foundations and Trends in Signal Processing, vol. 17, no. 4, 2023.

[7] Course lecture notes 1–8, Model-Based Deep Learning (361.2.2320), Ben-Gurion University, 2026.

Thank you very much for listening!